



Mechanical and physical properties of binderless particleboard made from oil palm empty fruit bunch (OPEFB) with addition of natural binder

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ABSTRACT

Malaysia produces a very high amount of oil palm biomass especially the oil palm empty fruit bunch (OPEFB) every year. The OPEFB in reality can be converted into value-added panels such as particleboard. The application may vary such as use for furniture, indoor building materials, door, window, floor, office partition and many more. Normally, the commercial particleboard were made from biomass with addition of formaldehyde based resins and it is harmful to the environment due to the formaldehyde released. In this study, the formaldehyde based resins were excluded and the binderless particleboard made from OPEFB with addition of natural binder such as corn, potato, tapioca and wheat starch were evaluated. These panels were manufactured with a target density of 0.8 g/cm³. The hot pressing temperature, hot pressing time, and pressing pressure used were 180 °C, 20 min, and 12 MPa respectively. From the results, the highest modulus of rupture (MOR) and modulus of elasticity (MOE) were found to be the panel with the addition of potato starch as the natural binder (15.33 MPa, 2078 MPa) following by wheat (14.41 MPa, 1823 MPa), tapioca (14.18 MPa, 1771 MPa) and corn starch (12.59 MPa, 1335 MPa) respectively. The highest dimensional stability was also found to be the panel with the addition of potato starch with the lowest water absorption (69%) and thickness swelling (50%) value. Overall results has met the Japanese Industrial Standard (JIS) for type-8 and type-13 except for water absorption and thickness swelling. Therefore, particleboard made from OPEFB without using any synthetic resin was successfully made with the addition of natural binder such as potato, wheat, tapioca and corn starch and probably can be replace as a substitute for conventional particleboard in future.

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1. Introduction

Oil palm (*Elaeis guineensis*) is one of the biggest Malaysia valuable produced products that helps in aspects of economic

and agricultural [1]. Malaysian has produced about partly of the world palm oil which is about 10.8 million tonnes ever year. Thus, it make Malaysia is the second largest producer and exporter of oil palm during this term than other country after Indonesia [2]. Oil palm empty fruit bunch (OPEFB) is classified as waste of oil palm industry which is produced from the oil extraction mills process [3]. From oil palm industries that also producing lignocelluloses waste such as mesocarp fruit fibres (MF), palm kernel shells (PKS), oil palm fronds (OPF), oil palm trunks (OPT), palm shells

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